

CSE 400: Fundamentals of Probability in Computing

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Lecture: L11_12: Transformation of Random Variables

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1 Transformation of a Single Random Variable

1.1 Definitions and Notation

- **Initial RV:** X with a known PDF $f_X(x)$ and CDF $F_X(x)$.
- **Transformation Function:** $g(x)$.
- **New RV:** $Y = g(X)$.
- **Goal:** Find the PDF $f_Y(y)$ and CDF $F_Y(y)$.

1.2 Assumptions / Conditions

- The function $g(x)$ is typically assumed to be monotonic (e.g., $\log(x)$).
- The inverse function $x = g^{-1}(y)$ must exist to solve for X in terms of Y .

1.3 Main Results / Theorems

The PDF of the transformed variable Y is given by:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Alternatively expressed as:

$$f_Y(y) = \frac{f_X(x)}{\left| \frac{dy}{dx} \right|} \Big|_{x=g^{-1}(y)}$$

1.4 Proofs / Derivations

- **Step 1:** Find the CDF of Y : $F_Y(y) = P(Y \leq y)$.
- **Step 2:** This simplifies to $P(g(X) \leq y) = P(X \leq g^{-1}(y))$.
- **Step 3:** Differentiate the CDF with respect to y using the chain rule:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y)) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

- **Step 4: Note:** If $g(x)$ is decreasing, $P(g(X) \leq y) = P(X \geq g^{-1}(y))$, leading to the absolute value in the general formula.

1.5 Worked Examples

Example 1: Uniform to Sine Transformation

- **Given:** $X \sim U(0, \pi/2)$, so $f_X(x) = \frac{2}{\pi}$ for $0 < x < \pi/2$.
 - **Transformation:** $Y = \sin(X)$.
 - **Step 1 (Inverse):** $x = \arcsin(y)$.
 - **Step 2 (Derivative):** $\frac{dx}{dy} = \frac{1}{\sqrt{1-y^2}}$.
 - **Step 3 (PDF):** $f_Y(y) = \frac{2}{\pi\sqrt{1-y^2}}$ for $0 < y < 1$.
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2 Function of Two Random Variables ($Z = X + Y$)

2.1 Definitions and Notation

- **Input RVs:** X and Y with joint PDF $f_{X,Y}(x, y)$.
- **Transformation:** $Z = X + Y$.
- **Resulting PDF:** $f_Z(z)$.

2.2 Main Results / Theorems

If X and Y are **independent**, the PDF of Z is the **convolution** of their individual PDFs:

$$f_Z(z) = f_X(x) * f_Y(y) = \int_{-\infty}^{\infty} f_X(w) f_Y(z - w) dw$$

2.3 Proofs / Derivations

- **Step 1:** Write the CDF $F_Z(z) = P(X + Y \leq z)$.
- **Step 2:** Express as a double integral: $\int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y) dy dx$.
- **Step 3:** Differentiate with respect to z using the **Leibnitz Rule**:

$$f_Z(z) = \frac{d}{dz} \int_{-\infty}^{\infty} F_Y(z - x) f_X(x) dx$$

- **Step 4:** Apply Leibnitz to get $f_Z(z) = \int_{-\infty}^{\infty} f_Y(z - x) f_X(x) dx$.

2.4 Worked Examples

Example 1: Sum of Independent Normal RVs

- **Given:** $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ are independent.
- **Result:** $Z \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Example 2: Sum of Independent Exponential RVs

- **Given:** $f_X(x) = \lambda e^{-\lambda x}$ and $f_Y(y) = \lambda e^{-\lambda y}$ for $x, y \geq 0$.
 - **Result:** $f_Z(z) = \lambda^2 z e^{-\lambda z}$ for $z \geq 0$ (Erlang distribution).
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3 Other Joint Transformations ($X - Y$ and X/Y)

3.1 Definitions and Notation

- **Difference:** $Z = X - Y$.
- **Ratio:** $Z = X/Y$.

3.2 Proofs / Derivations

For $Z = X - Y$:

- **Step 1:** $F_Z(z) = P(X - Y \leq z) = P(X \leq z + Y)$.
- **Step 2:** $f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(z + y, y) dy$.

For $Z = X/Y$:

- **Step 1:** $F_Z(z) = P(X/Y \leq z)$.
- **Step 2:** Rewrite as $P(X \leq zY)$ (assuming $Y > 0$).
- **Result:** $f_Z(z) = \int_{-\infty}^{\infty} |y| f_{X,Y}(zy, y) dy$.

3.3 Worked Examples

Example 1: Magnitude of Two RVs

- **Find:** PDF of $Z = \sqrt{X^2 + Y^2}$ (often leading to Rayleigh distribution if X, Y are i.i.d. Normal).